DATABASE LAB @POSTECH

Knowledge-augmented Language Model Prompting for Zero-shot Knowledge Graph Question Answering

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Recap

Retrieval Augmented Generation

Motivation: Traditional language models rely only on *parametric knowledges* which are

- hard to fine-tune
- hard to explain (e.g., why is this output generated?)

Idea: *Retrieve*, then *generate*

- A *retriever* retrieves query-relevant documents from a corpus (i.e., database containing text)
- A *reader* generates the output based on the query and the retrieved results

Background

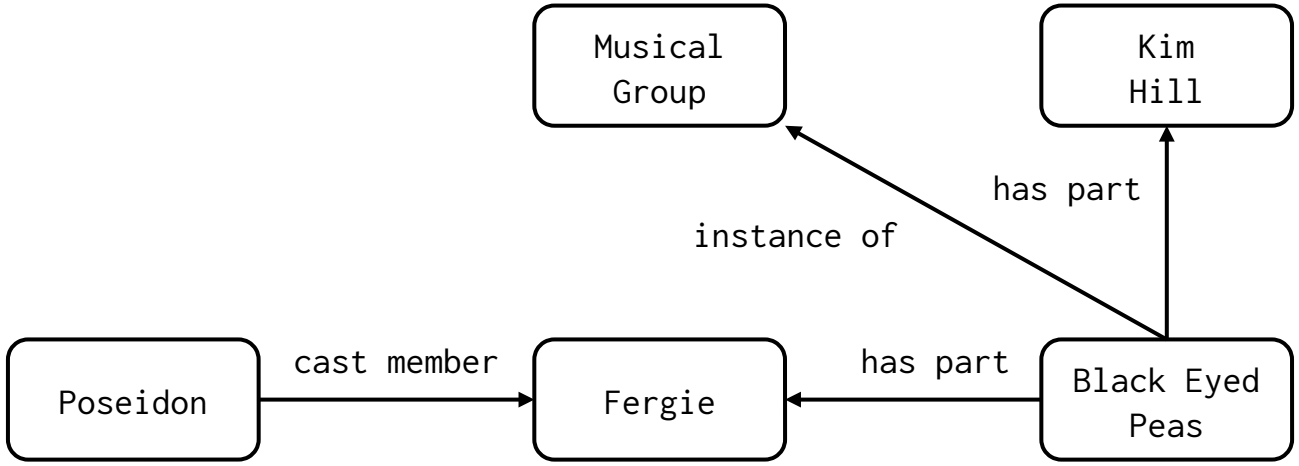
Knowledge Graph

A knowledge graph (KG) is a node-labeled, edge-labeled graph where

- a node represents a real-world entity (objects, events, situations or concepts)
- an edge represents a relationship between two entities

A knowledge graph can also be viewed as a set of triples

- Triple is a 3-tuple that contains two entities and a relation between them



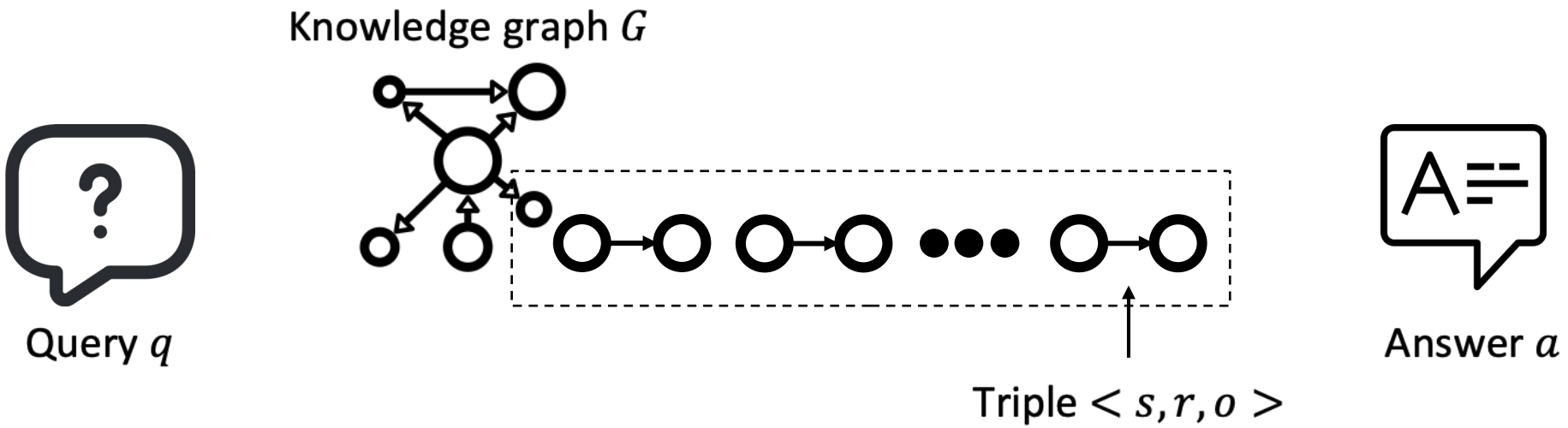
An example knowledge graph

```
< Poseidon, cast member, Fergie >
< Black Eyed Peas, has part, Kim Hill >
< Black Eyed Peas, has part, Fergie >
< Black Eyed Peas, instance of, Musical Group >
```

An set of triples equivalent to the left KG

Problem Description

- Given a query q and a knowledge graph G, generate the correct answer a.
- Query $q = [q_1, \ldots, q_t]$: A sequence of tokens
- Knowledge graph G: A set of triples
  - Triple $< s, r, o >$: A tuple of subject (entity) – relation – object (entity)
- Answer $a = [a_1, \ldots, a_r]$: A sequence of tokens

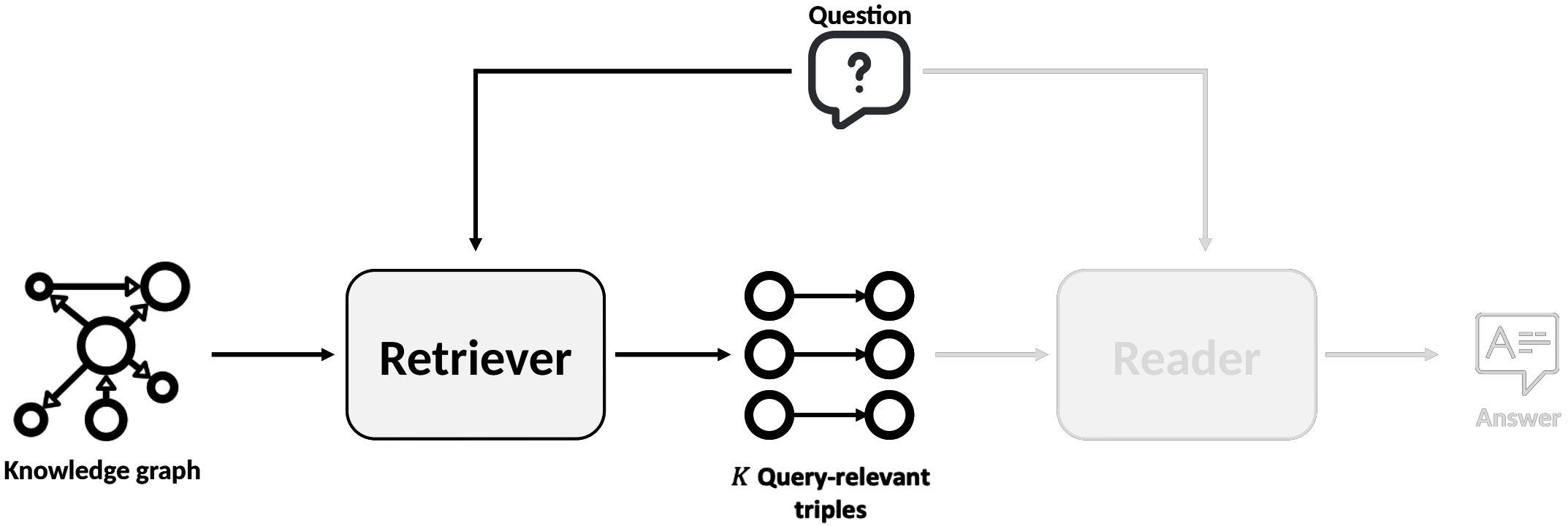


Problems of Existing Works

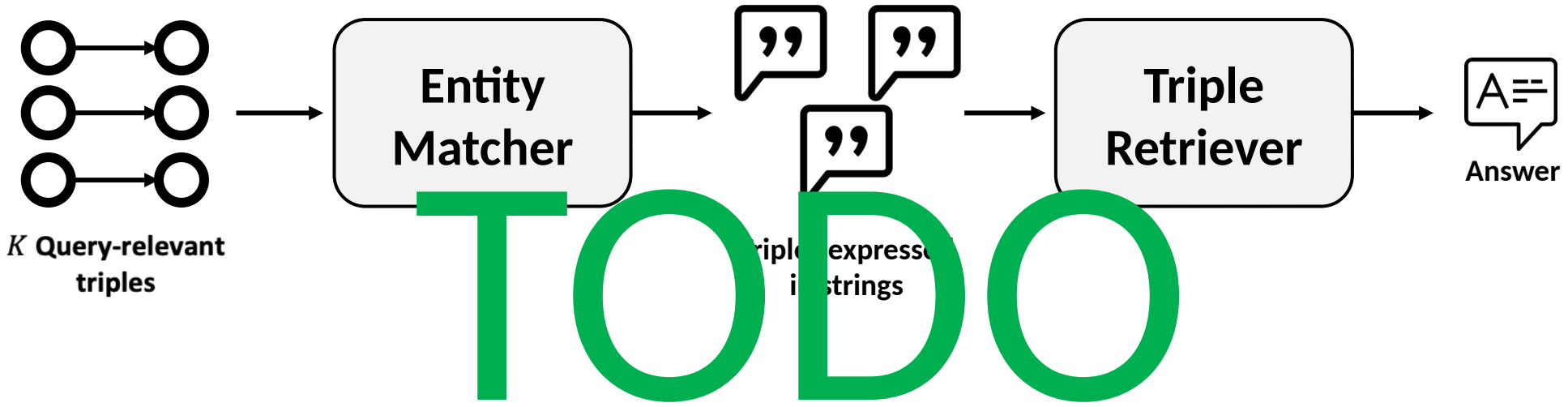
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KAPING Overview

KAPING has a very trivial RAG form.



KAPING: Retriever



KAPING: Retriever – Entity Matching

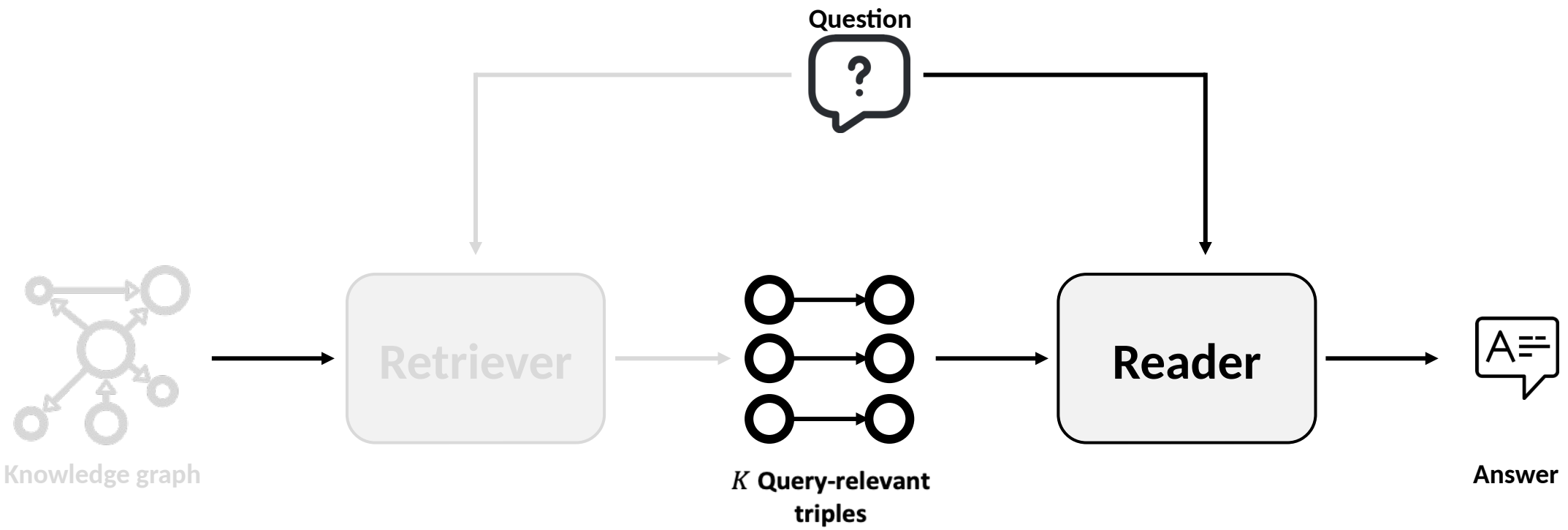
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KAPING: Retriever – Triple Retrieving

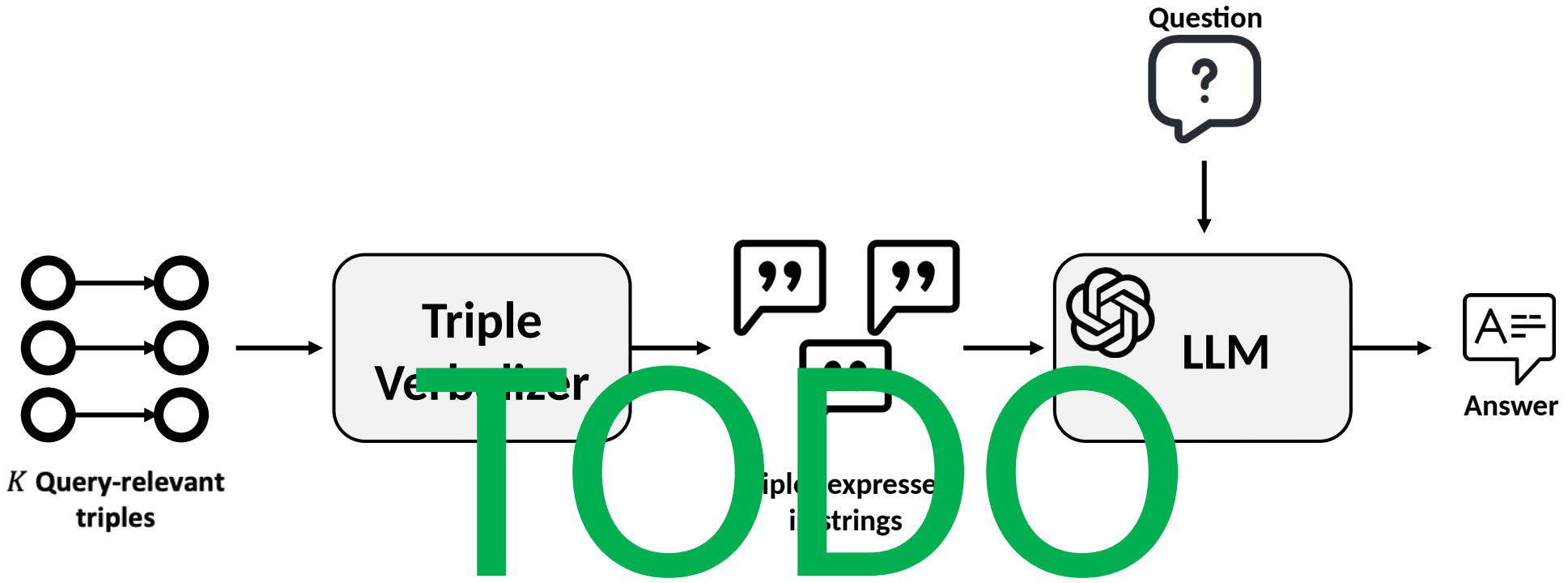
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KAPING Overview (Revisited)

KAPING has a very trivial RAG form.



KAPING: Reader



KAPING: Reader – Prompt Generation

TODO

Limitations of KAPING

1. KAPING relies on the accuracy of on-the-shelf entity matching algorithm.
2. KAPING can only find evidences of limited forms.
3. KAPING's verbalization scheme is very naïve.

Limitation 1: Relies on Entity Matching Algorithm

TODO

Limitation 2: Only Finds Evidences of Limited Forms

TODO

Limitation 3: Naïve Verbalization Scheme

TODO

Experiments

Used Metrics

End-to-end accuracy

- Substring match: Does any substring of the predicted answer contain any of the set of ground truth answers?

< Example 1 >
Predicted answer: "The answer is dog."
Ground truth answers: {dog, canine, doggy}
Substring match accuracy: 1

< Example 2 >
Predicted answer: "He likes phones."
Ground truth answers: {computer, PC}
Substring match accuracy: 0

Retriever accuracy

- MRR (Mean reciprocal rank)
- Hit@k

Experiment Settings

Datasets

- WebQuestionsSP: based on FreebaseKG
- Mintaka: based on Wikidata KG

Baselines: variants of KAPING based on how to set the retrieved triples

- `NoKnowledge`: No triples
- `RandomKnowledge`: Randomly sampled triples
- `PopularKnowledge`: Triples containing the most frequently occurring relations
- `GeneratedKnowledge`: LLM-extracted parametric knowledge

Experiment 1: End-to-end Accuracy Comparison

Table 1: **Main results of language model prompting**, where we report the generation accuracy. The number inside the parentheses in the first row denotes the parameter size of language models, and best scores are emphasized in bold.

Datasets	Methods	T5 (0.8B)	T5 (3B)	T5 (11B)	OPT (2.7B)	OPT (6.7B)	OPT (13B)	T0 (3B)	T0 (11B)	GPT-3 (6.7B)	GPT-3 (175B)	AlexaTM (20B)	Average
WebQSP w/ Freebase	No Knowledge	6.95	13.40	9.48	19.85	29.77	28.38	21.43	40.77	44.63	63.59	46.79	29.55
	Random Knowledge	21.55	19.15	17.57	28.07	31.73	33.31	32.62	51.20	51.01	65.87	57.37	37.22
	Popular Knowledge	15.30	16.88	18.39	28.32	28.13	24.21	27.05	47.22	45.58	62.26	54.91	33.48
	Generated Knowledge	6.19	7.84	6.76	7.46	11.50	8.22	19.41	38.81	45.89	62.14	35.13	22.67
	KAPING (Ours)	**34.70**	**25.41**	**24.91**	**41.09**	**43.93**	**40.20**	**52.28**	**62.85**	**60.37**	**73.89**	**67.67**	**47.94**
WebQSP w/ Wikidata	No Knowledge	10.30	18.42	15.21	23.94	33.77	32.40	24.56	44.20	48.50	67.60	42.41	32.85
	Random Knowledge	17.94	22.78	24.28	37.24	35.61	38.27	28.85	47.68	52.05	60.64	55.63	38.27
	Popular Knowledge	15.35	20.80	20.74	30.83	30.01	27.83	24.83	48.02	47.41	63.37	53.92	34.83
	Generated Knowledge	11.94	13.30	12.28	11.26	17.53	14.19	22.92	41.34	48.77	65.89	31.16	26.42
	KAPING (Ours)	**23.67**	**40.38**	**35.47**	**49.52**	**53.34**	**51.57**	**49.86**	**58.73**	**60.44**	**69.58**	**65.04**	**50.69**
Mintaka w/ Wikidata	No Knowledge	11.23	14.25	17.06	19.76	27.19	26.83	14.75	23.74	34.65	56.33	41.97	26.16
	Random Knowledge	17.59	18.19	18.83	28.11	26.58	28.36	16.10	26.15	32.98	51.56	46.02	28.22
	Popular Knowledge	17.56	18.09	18.73	26.97	27.08	23.10	16.74	27.15	32.48	53.16	46.41	27.95
	Generated Knowledge	13.61	14.61	14.29	11.87	14.96	16.24	14.46	23.13	33.12	55.65	34.58	22.41
	KAPING (Ours)	**19.72**	**22.00**	**22.85**	**32.94**	**32.37**	**33.37**	**20.68**	**29.50**	**35.61**	**56.86**	**49.08**	**32.27**

- KAPING outperforms every other baseline models.
- RandomKnowledge and PopularKnowledge mostly outperform the NoKnowledge model.
  - "These augmented knowledges are sometimes useful to answer the question."
- GeneratedKnowledge model mostly underperforms NoKnowledge model.
  - The LLM-extracted parametric knowledge incurs more hallucination.

Experiment 2: Retriever Accuracy Comparison

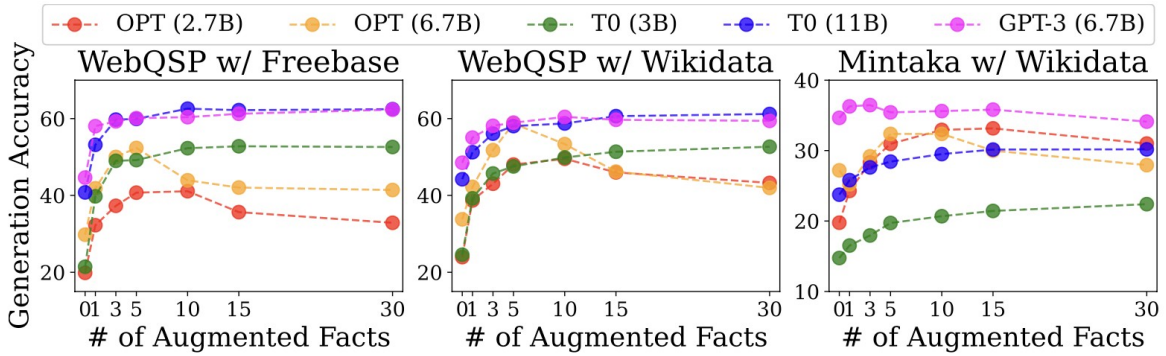
Datasets	Retrievers	1-Hop Retrieval				2-Hop Retrieval			
		MRR	Top-1	Top-10	Top-30	MRR	Top-1	Top-10	Top-30
WebQSP w/ Freebase	Random	12.50	7.21	25.09	34.64	1.50	0.70	2.65	5.37
	Popular	8.58	5.31	15.93	24.53	1.59	0.95	2.72	4.68
	MPNet	**47.27**	**40.27**	**60.56**	**64.48**	**41.64**	**33.12**	**58.47**	**65.23**
WebQSP w/ Wikidata	Random	9.50	3.62	22.58	40.72	1.31	0.00	2.80	8.59
	Popular	8.52	4.57	15.89	35.47	4.63	4.02	5.53	6.62
	MPNet	**43.46**	**33.36**	**64.39**	**70.67**	**40.42**	**30.56**	**62.62**	**71.56**
Mintaka w/ Wikidata	Random	4.80	1.85	11.48	22.03	0.91	0.14	1.78	5.15
	Popular	6.09	3.09	12.51	20.47	0.24	0.04	0.28	1.24
	MPNet	**13.01**	**7.50**	**25.44**	**35.43**	**13.00**	**6.82**	**26.65**	**40.01**

Table 2: **Retriever results.** We compare random model, popular model, and MPNet (Song et al., 2020), on 1- and 2-hop retrievals.

- Retrieval using MPNet shows superior accuracy against naïve models.
- "When we increase the hop-size from one to two, the retriever is more likely to retrieve irrelevant triples that does not include answer entities".

Experiment 3: Parameter Sensitivity

Figure 5: **Performances with varying knowledge amount**, where we change the number of retrieved triples to augment.



Models	# of Retrieved Facts	Relative Time T0 (3B)	Relative Time OPT (2.7B)
No Knowledge	0	1.00	1.00
KAPING (Ours)	1	0.49	1.12
	5	0.73	1.48
	10	1.07	1.89
	15	1.54	2.36
	30	2.49	3.77

Table 3: **Efficiencies with varying the knowledge amount**, where we measure the wall-clock time of every model for generating the answer on the WebQSP w/ Wikidata dataset.

- End-to-end accuracy and algorithm runtime comparison on varying numbers of retrieved triples K
  - The end-to-end accuracy is the highest when K reaches about 5 or 10.
  - The end to-end accuracy diminishes when K exceeds 15 or 30, due to hard negatives.
  - The algorithm runtime for KAPING ($K < 10$) is shorter than NoKnowledge
    - The decoding time of KAPING is shorter, as the LLM tends to generate shorter answers when the retrieved results are given.

Limitations of KAPING's Experiment

- The authors did not compare KAPING with other KGQA methods.
- Insufficient analysis on some experiment results
  - Ex1) Why doesn't the end-to-end accuracy of GPT and T0 diminish as K increase?